LSP Task - 6

Privilege levels in Linux : kernel mode and user mode

Linux, like most modern operating systems, uses privilage levels to protect system resources and ensure stable, secure operation. These Privilage levels dictate what a process can and cannot do. Linux privilage mode primarily operates with two privilage levels: kernel mode and user mode.

Privilage levels

Privilage levels are part of the CPU protection mechanism. In x86 architecture, there are four privilage rings (0 to 3), with Ring 0 having the most privilage and Ring 3 the least.

- kernel mode (Ring 0)
- user mode (Ring 3)

Kernel mode

code running in this mode can :

- Access any memory address
- Interact directly with hardware (eg-disk drives)
- Execute privilaged CPU instructions
- control the scheduling of processes and threads

Operations permited in kernel mode include :

- Managing memory: Allocating and deallocating memory for processes.
- process control : creating and terminating processes.
- I/O operations : Reading and writing to devices
- system calls execution : Handling requests from user mode .

example :

when a process calls the open() system call to access a file , the call transitions to kernel mode where the kernel checks file permissions and manages the file descriptors .

User Mode

- User mode is less privileged level, where most user applications run. In this mode, code can't directly access hardware or kernel memory, preventing accidental or malicious damage to this system.

Operation permitted in user mode:

- Running user applications like next editors, browsers and compilers.
- performing computations and logic operations.
- making system calls request kernel services.

Example

A text editor like vim operates in user mode. when the user saves a file, vim uses system calls to ask the kernel to write data to disk, but it can't directly access the disk hardware.